LAKSHMANAN L

+91 9025298998 | lakshmananloganathan317@gmail.com | linkedin/lakshmanan-loganathan | github/Lakshmanan-04

Education

SRM Institute of Science and Technology (KTR) **CGPA: 9.45/10**
B.Tech in Computer Science and Engineering (Core) *2023 – 2027*

Internship

Internship: Writer Information **Chennai, Tamil Nadu**
Associate HR Intern – Human Resource Development *Dec 2025*

- Assisted with HR operations, including maintaining employee records, updating documentation, and organizing personnel information.
- Supported recruitment and talent acquisition activities through interview scheduling, candidate communication, resume screening, and coordination with internal stakeholders.
- Contributed to HR documentation and administrative activities to improve record accuracy and streamline workflows.
- Gained practical exposure to talent acquisition, employee lifecycle processes, and human resource management practices.

Projects

Machine Learning Based EV Battery Health Prediction | *Gradient Boosting — Python, Scikit-learn, Pandas, NumPy*

- Developed an EV Battery Health Prediction system using Python and machine learning to estimate battery State of Health (SOH) and support battery performance analysis.
- Implemented and evaluated machine learning models, including Gradient Boosting, to predict battery health based on relevant battery performance parameters..
- Analyzed model performance using evaluation metrics to assess prediction accuracy and demonstrate the potential for predictive maintenance and improved EV battery reliability.

Sign Language to Text Conversion System | *TensorFlow, Python, CNN, FastAPI*

- Developed a sign language recognition system using Python and deep learning to identify hand gestures from images or video frames and translate them into readable text.
- Implemented and trained a Convolutional Neural Network (CNN) to classify sign language gestures and convert recognized signs into corresponding text outputs
- Built an interactive user interface to display predicted signs and their text interpretations, integrating the machine learning model with a backend for efficient prediction and user interaction.

Integrated Animal Shelter Management Platform | *Java, JDBC, MySQL, SQL, OOPS*

- Developed a platform to streamline animal shelter operations by managing animal profiles, shelter records, and adoption details.
- Implemented JDBC connectivity to integrate the Java application with a MySQL database for storing, retrieving, and updating animal information.
- Applied object-oriented programming principles and SQL queries to organize animal records, track adoption status, and maintain data consistency.

Air Quality Monitoring System | *WSN, IoT, ESP32, Sensors, ThingSpeak*

- Designed an IoT-based air quality monitoring system using wireless sensor networks (WSN) and ESP32 to collect and monitor environmental parameters for real-time air quality assessment.
- Implemented sensor data collection and wireless communication, with ThingSpeak-based visualization to monitor air quality readings and analyze environmental trends.

Technical Skills

Languages: Python, Java, C, C++, SQL
ML/DL: PyTorch, TensorFlow, Scikit-learn, CNN, LSTM, ViT, RNN, Transfer Learning, NLP, Prompt Engineering
Web & Backend: React.js, Node.js, FastAPI, JDBC, HTML/CSS/JavaScript
Tools & Cloud: GitHub, AWS, MongoDB, Firebase, OpenCV, Docker
Core Areas: Data Structures & Algorithms, OOPS, DBMS, Machine Learning, Cloud Computing, Software Engineering

Certifications

- AWS Certified Developer – Associate
- Microsoft Certified: Azure Developer Associate (AZ 204)
- NPTEL Certificate in Programming in JAVA
- NPTEL Certificate in Machine Learning